

Technical Supplement

Estimating Factor Loadings with PCA

The Goal: Recovering unobserved factor loadings

Recall the firm-specific labor demand equation:

$$L_{it} = \phi^d w_t + \lambda_i \eta_t + u_{it}$$

- **The Challenge:** We need weights Γ orthogonal to factor loadings ($\sum \Gamma_i \lambda_i = 0$) to isolate idiosyncratic shocks. However, λ_i (macro-sensitivity) is **unobserved**.
- **The Solution:** If idiosyncratic shocks u_{it} are small, the dominant variance in labor demand L_{it} is driven by the macro shock η_t .
- **The Method:** We can use **Principal Component Analysis (PCA)** on the panel of labor outcomes to extract the largest source of variance.

Step 1: The Residual Matrix and Covariance

Assume we partial out the wage effect $\phi^d w_t$. We are left with the residual labor variation driven by shocks:

$$Y_{it} = \lambda_i \eta_t + u_{it}$$

Let Y be the $T \times N$ matrix of these demeaned observations. The sample **covariance matrix** across firms is:

$$\Sigma = \frac{1}{T} Y' Y$$

Because firms are primarily correlated through the common macro shock η_t , the matrix Σ contains the structural signal of λ .

Step 2: Formulating the Optimization Problem

PCA seeks a vector of weights $v \in \mathbb{R}^N$ that **maximizes variance**.

The variance of the data projected onto v is proportional to $v'\Sigma v$. To prevent infinite variance, we constrain the length of v to 1.

The Constrained Optimization Problem:

$$\begin{aligned} \max_v \quad & v'\Sigma v \\ \text{subject to} \quad & v'v = 1 \end{aligned}$$

Step 3: Solving via Lagrange Multipliers

We construct the Lagrangian $\mathcal{L}$ with a multiplier α for the constraint $v'v - 1 = 0$:

$$\mathcal{L}(v, \alpha) = v'\Sigma v - \alpha(v'v - 1)$$

Taking the derivative with respect to the vector v and setting it to zero:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial v} &= 2\Sigma v - 2\alpha v = 0 \\ \Sigma v &= \alpha v\end{aligned}$$

This is the fundamental **Eigenvalue Equation**. The solution v is an eigenvector of the covariance matrix Σ , and α is its eigenvalue.

Step 4: Eigenvalues measure captured variance

What does the eigenvalue α represent economically? Pre-multiply the eigenvalue equation by v' :

$$\Sigma v = \alpha v$$

$$v' \Sigma v = v' (\alpha v)$$

$$v' \Sigma v = \alpha (v' v)$$

$$v' \Sigma v = \alpha \quad (\text{since } v' v = 1)$$

The eigenvalue α is **exactly equal** to the variance captured by the eigenvector v . Since we want to maximize variance, we select the eigenvector v_1 associated with the **largest** eigenvalue α_1 .

Step 5: Recovering the GIV Weights

The principal eigenvector v_1 captures the dominant variance, which in our model is the macro shock η_t . Therefore, v_1 is our empirical estimate of the factor loadings:

$$\hat{\lambda} = v_1$$

With $\hat{\lambda}$ recovered, we can now construct the optimal Granular Instrument:

1. Compute the orthogonal projection matrix: $\hat{Q} = I - \hat{\lambda}(\hat{\lambda}'\hat{\lambda})^{-1}\hat{\lambda}'$
2. Project the size weights: $\hat{\Gamma}^* = S'\hat{Q}$
3. Construct the GIV: $z_t = \sum \hat{\Gamma}_i^* L_{it}$